

**IBM DATA SCIENCE PROFESSIONAL CERTIFICATE — CAPSTONE
PROJECT**

SpaceX Falcon 9 First Stage

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Question 1

Git hub Repository

<https://github.com/JuanSer10/datasciencecoursera>

Repository contents:

- `jupyter-labs-spacex-data-collection-api.ipynb`
- `labs-jupyter-spacex-Data_wrangling.ipynb`
- `05_SpaceX_EDA_Data_Visualization.ipynb`
- `spacex-dash-app.py`
- `jupyter-labs-eda-sql-coursera_sqlite.ipynb`
- `06_SpaceX_Interactive_Visual_Analytics_Folium.ipynb`
- `SpaceX-Machine-Learning-Prediction.ipynb`

OUTLINE

Execute summary

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Executive summary

- Summary of methodologies
- Data Collection
- Data Wrangling
- Exploratory Data Analysis with Data Visualization
- Exploratory Data Analysis with SQL
- Building an interactive map with Folium
- Building an Dashboard with plotly Dash
- Predictive analysis
- Summary of all results
- Exploratory Data Analysis results
- Predictive analysis Results

INTRODUCCION

Welcome to the Applied Data Science Capstone Project, where we embark on a journey to predict the successful landing of the Falcon 9 first stage. SpaceX, known for revolutionizing space travel, advertises Falcon 9 rocket launches at a competitive cost of 62 million dollars, compared to other providers' costs upwards of 165 million dollars. This significant cost-saving is largely attributed to SpaceX's ability to reuse the first stage of the rocket. By accurately predicting the landing success of the first stage, we can better estimate the launch costs, providing valuable insights for companies bidding against SpaceX for rocket launches.

Data collection and Data Wrangling methodology

The project is structured into several comprehensive modules, each designed to build on the previous one, culminating in a robust predictive model. Below is a detailed description of the tasks and goals achieved in each module.

GITHUB URL:<https://github.com/JuanSer10/datasciencecoursera/tree/main>

Data collection and Data Wrangling methodology

Data collection process involved a combination of API requests from spaceX REST API and web scrabbing data from a table in spaceX is Wikipedia entry

We had to use both of these Data collection methods in order to get complete information about the launches for a more detailed analysis

Data Columns are obtained by using SpaceX REST API

Flight number	Data	booster version	pay load mass
Orbit	launchsite	outcome	flights
Grid fins	Legs	landing pad	block
Reused count	serial	longitude	latitude

Data columns are obtained using Wikipedia

Flight No	Launch site	Payload	Pay Load Mass
Orbit	Cus tomer	launch outcome	Version Booster
Date	Time		

Data collection – SpaceX API

- Initiate API Request
- Use Python requests library to connect to the spaceX API
- Endpoint <https://api.spacexdata.com/v4/rockets/>
- Parse API Response
- Convert response from Json to Python
- Extract relevant fields: Launch date, launch site Payload mass Rocket type
- Save extracted data into a pandas DataFrame
- Store the DataFrame Locally For Further processing
- Replacing missing values of payload mass column with calculated
- Exporting the data to CSV

Data Wrangling

Cleaning



Transforming



Organizing

Step 1 Data cleaning

Identify and fill or remove missing values in the data set
Use techniques to deal with missing data

Step 2 Data Transformation

Convert data types to appropriate formats
Create new features
Normalize / scale numerical data

Step 3 Data integration

Merge datasets Collected from different sources
Consists column

Step 4 Data validation

GITHUB:

DATA WRANGLING

METHODOLOGY

In the data set there are several different cases where the booster did not land successfully

True RTLS means the mission outcome was successfully landed to ground pad.

False RTLS Means the mission outcomes was unsuccessfully landed to a ground pad

True ASDS means the mission outcome was successfully landed on a drone ship false

False ASDS means the mission outcome was unsuccessfully landed on a drone ship

Eda AND Data visualization methodology

➤ Aggregate queries

Calculated total number of launches

Calculate success rates

Join queries

Joined tables to link Launch records

Combined datasets for comprehensive

➤ Filtering Queries

Filtered data to focus on specific launch

Applied conditions to extract launch

Subqueries

Nested queries to calculate derived metrics

Subqueries used to perform detailed analysis within larger datasets

GITHUB URL:

https://github.com/JuanSer10/datasciencecoursera/blob/main/3%20jupyter-labs-eda-sql-coursera_sqlite%20.ipynb

Eda WITH DATA VISUALIZATION METHODOLOGY

Exploratory Data Analysis (EDA) is an essential first step in the data science process. It involves analyzing and summarizing a dataset to understand its main characteristics, uncover underlying patterns, spot anomalies, and check assumptions.

Charts were plotted

Flight number vs Payload Mass

GITHUB:

<https://github.com/JuanSer10/datasciencecoursera/blob/main/4%20jupyter%20lab%20dataviz.ipynb>



► Histograms

What they do: Show the distribution of a single, continuous numerical variable by splitting the data into intervals (called "bins").

- **What they are used for:**

- Identifying the shape of the data (e.g., normal distribution, skewed left, skewed right).
- Finding where most values cluster.
- Spotting outliers or gaps in your data.
- *Example:* Visualizing the distribution of employee salaries or house prices.

► 2. Bar Charts

- **What they do:** Compare different categories or groups using rectangular bars, where the length of the bar represents the value.

- **What they are used for:**

- Comparing discrete categories (not continuous numbers).
- Tracking changes over time when the changes are large or grouped by year/month.
- *Example:* Comparing the total sales across different product categories (e.g., Electronics vs. Clothing vs. Groceries).



3. Line Charts

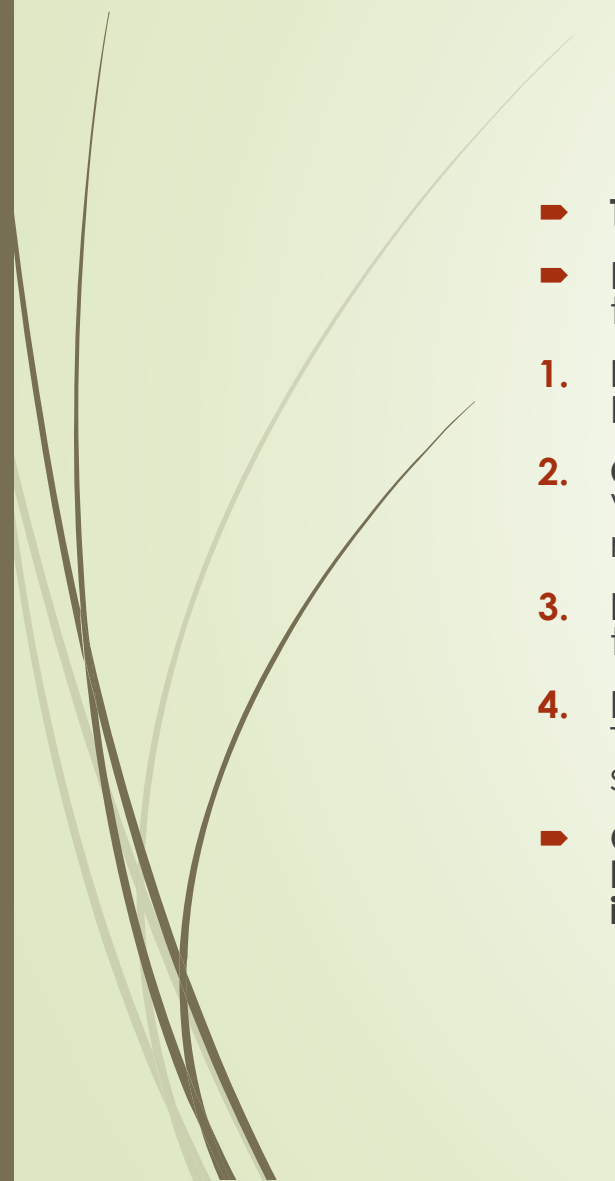
- **What they do:** Connect a series of individual data points with a continuous line, typically ordered chronologically.
- **What they are used for:**
 - Tracking trends, cycles, or changes over continuous time intervals.
 - Comparing performance or behavior of multiple groups over the same period.
 - *Example:* Looking at a company's monthly stock price fluctuations over five years.

4. Scatter Plots

- **What they do:** Display individual data points on a two-dimensional grid using horizontal (\$X\$) and vertical (\$Y\$) axes.
- **What they are used for:**
 - Determining the relationship or correlation between two continuous variables.
 - Checking if an increase in one variable leads to an increase (positive correlation) or decrease (negative correlation) in another.
 - Identifying clusters of data or extreme outliers.
 - *Example:* Plotting a person's height against their weight to see if a correlation exists.



Predictive analysis

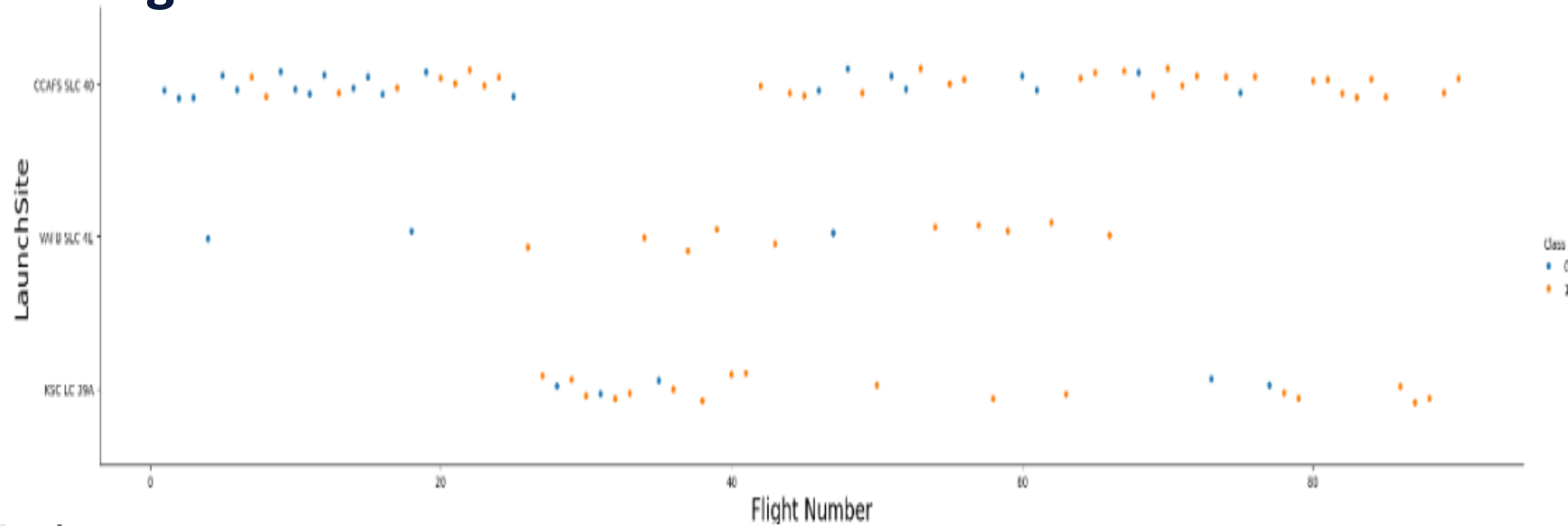


- **The Selection Process (How it's done)**

- Data scientists rarely just pick one model and hope for the best. They use a systematic process to test and select the winner:
 1. **Establish a Baseline:** Start with a very simple model (like a simple average or basic Linear Regression). Any complex model you build later *must* beat this baseline to be worth the effort.
 2. **Cross-Validation:** Split the historical data into multiple training and testing sets (e.g., K-Fold Cross-Validation). This ensures the model performs well on *unseen* data, preventing **overfitting** (where a model memorizes the training data but fails in the real world).
 3. **Hyperparameter Tuning:** Fine-tune the internal settings of the top-performing models (using techniques like Grid Search or Random Search) to squeeze out the best possible performance.
 4. **Ensemble Methods:** Often, the best model isn't a single algorithm, but a combination of them. Techniques like **Voting** or **Stacking** combine predictions from multiple models to create a stronger, more robust final prediction.
- **GITHUB:**
[https://github.com/JuanSer10/datasciencecoursera/blob/main/6%20SpaceX_Machine%20Learning%20Prediction_Part_5%20\(1\).ipynb](https://github.com/JuanSer10/datasciencecoursera/blob/main/6%20SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb)

EDA WITH VISUALIZATION

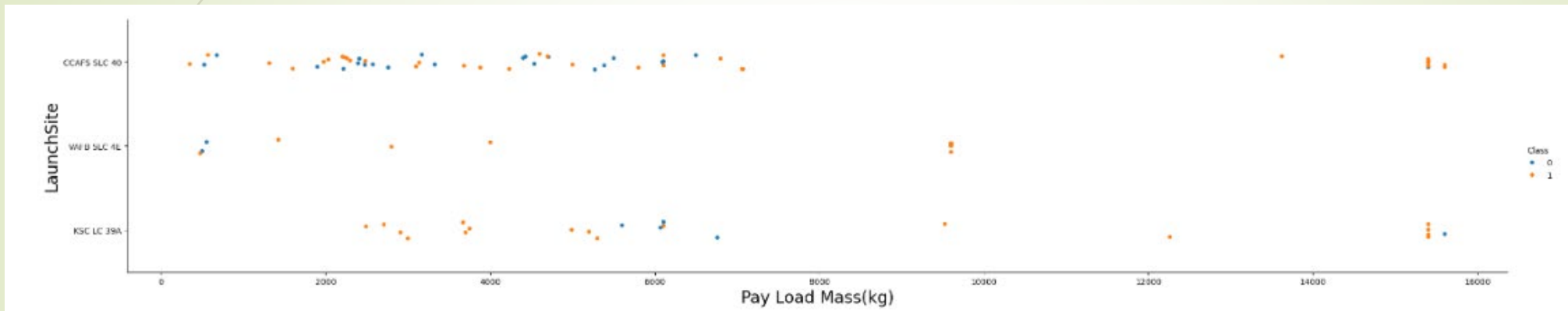
Flight Number vs. Launch Site



Early flights (low FlightNumber) skew toward failed landings (blue); later flights at every site trend orange (success) — experience improves outcomes

The CCAFS SLC 40 launch site has about a half of all launch
VAFB SCL 4E KSC LC 39 have a higher success rate

Payload Mass vs. Launch Site

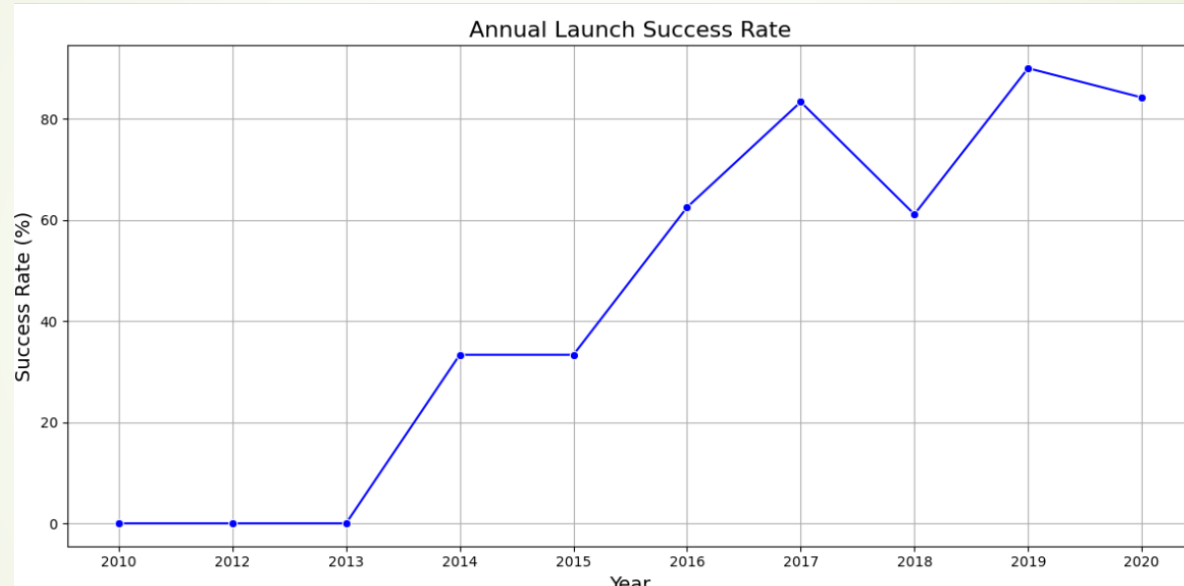


For every launch site higher the payload mass the higher success rate

Most of the lanches with payload mass over 7000 kg were successful

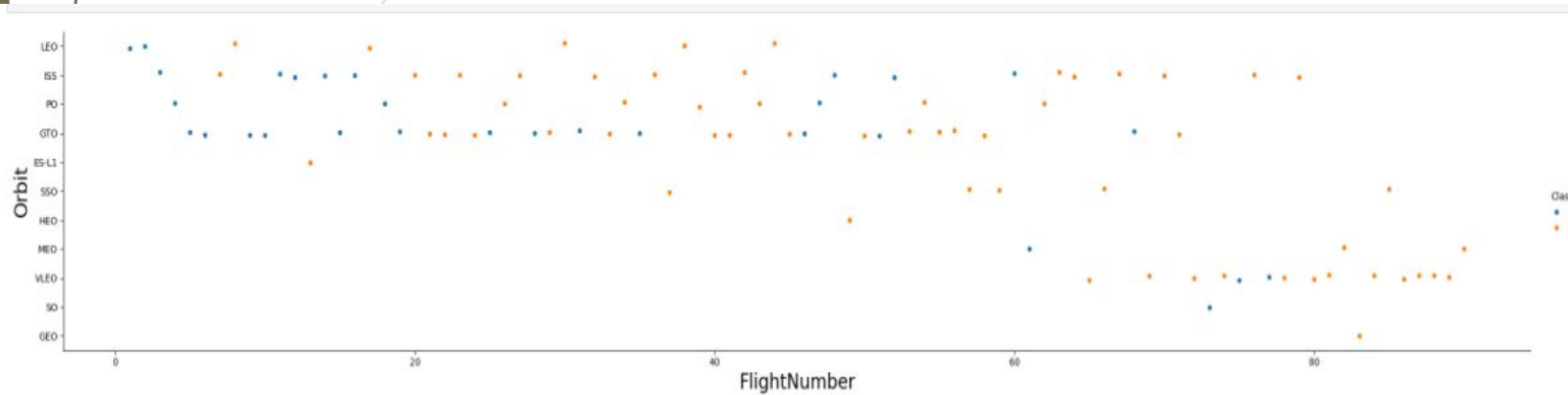
KSC LC-39A handles the heaviest payloads and still lands successfully most of the time; VAFB SLC-4E sees the fewest, lightest launches.

Annual launch success rate



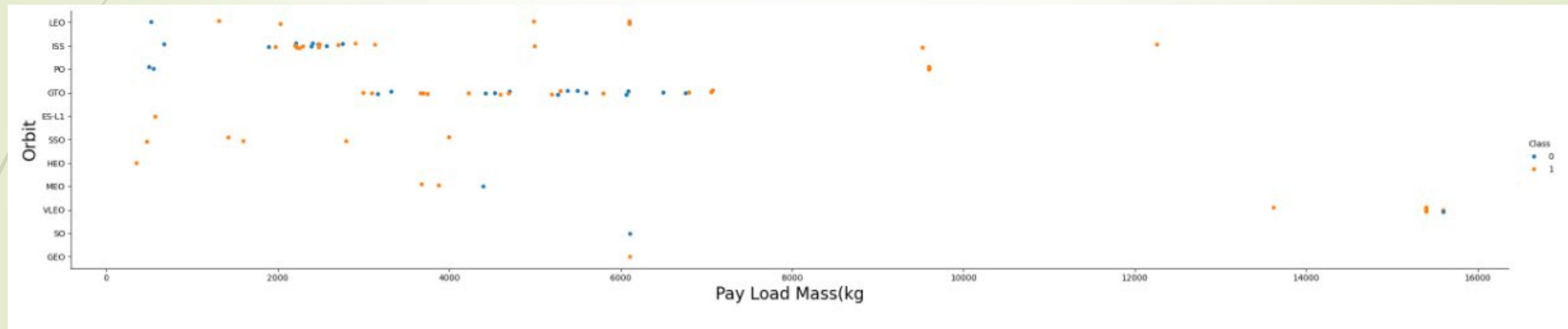
Success rate climbs year over year, from 0% in 2010–2012 to a sustained streak of 100% by 2018 onward.

Flight Number vs. Orbit



LEO, ISS, GTO missions span the whole timeline; VLEO missions only appear from flight ~60 onward and land almost every time..

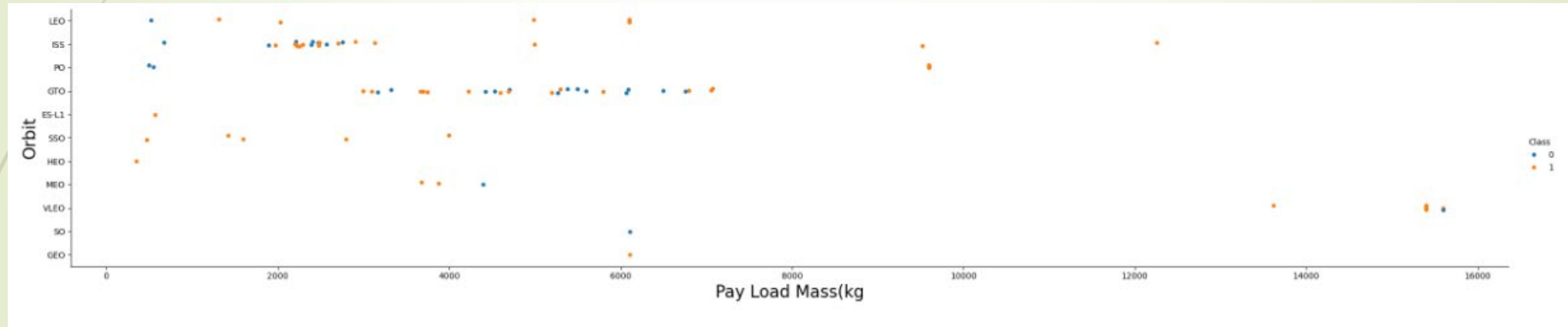
Payload Mass vs. Orbit



GTO missions carry the widest range of payload mass; the heaviest payloads of all (13,000–16,000 kg) go to VLEO and still land

Heavy payloads have a negative influence on GTO orbits and positive on GTO and Polar LEO iss orbits

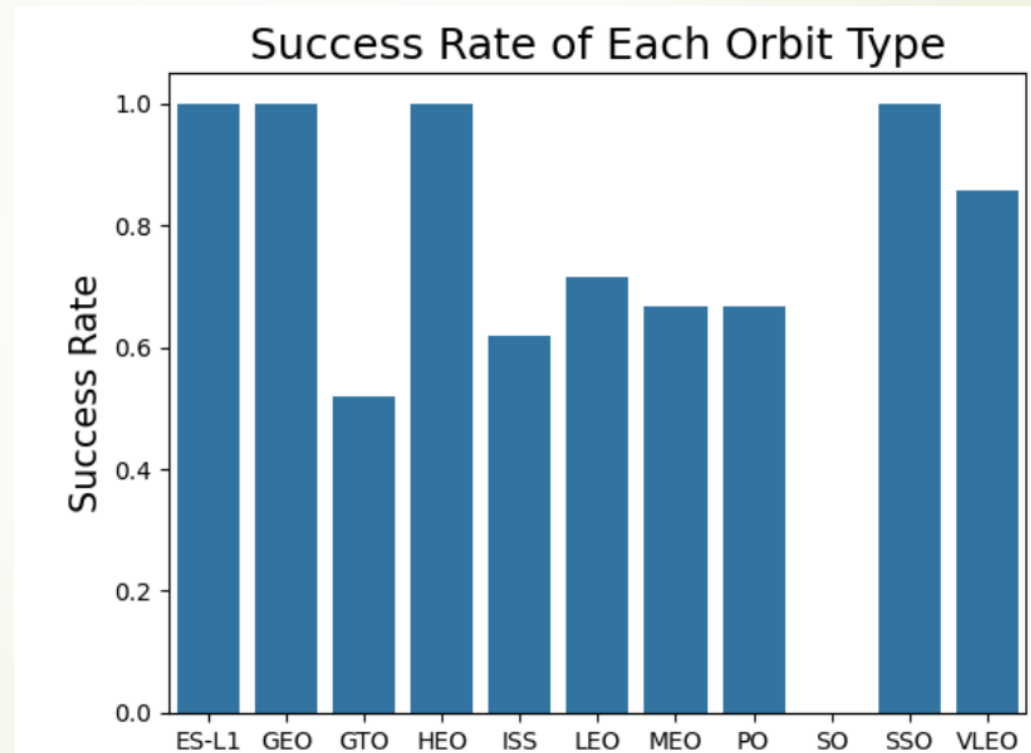
Payload Mass vs. Orbit



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SUCCESS RATE OF EACH ORBIT TYPE



ES-L1, GEO, HEO and SSO orbits are a perfect 100% in this dataset; GTO is the toughest at ~52%, and SO has zero recorded successes.

EDA WITH SQL

Launch site names

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT "Launch_site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db  
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Displaying the names of
the unique launch sites
in the space mission

TOTAL PAYLOAD MASS

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
] : %sql SELECT SUM("PAYLOAD_MASS__KG_") AS "TOTAL_PayloadMass" FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';  
  
* sqlite:///my_data1.db  
Done.  
]  
]: TOTAL_PayloadMass  
      45596
```

Displaying the total payload mass carried by
booster launched by NASA(CRS)

Launch site names begin with cca

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "launch_Site" LIKE 'CCA%' LIMIT 5;
```

* sqlite:///my_data1.db
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Displaying 5 records where launch sites begin with the string CCA

Average payload mass by f9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql select AVG("PAYLOAD_MASS_KG_") as Avg_payloadMass from SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'
```

```
* sqlite:///my_data1.db  
Done.
```

Avg_payloadMass

2928.4

Displaying average payload mass carried by booster f9 v1.1

First successful ground landing date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql Select MIN(Date) AS "FIRST Successfull landing" From SPACEXTABLE WHERE Landing_Outcome = 'success(ground pad)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
FIRST Successfull landing
```

```
None
```

Listing the date when the first successful landing outcome in ground pad was achieved

Task 6

List the names of

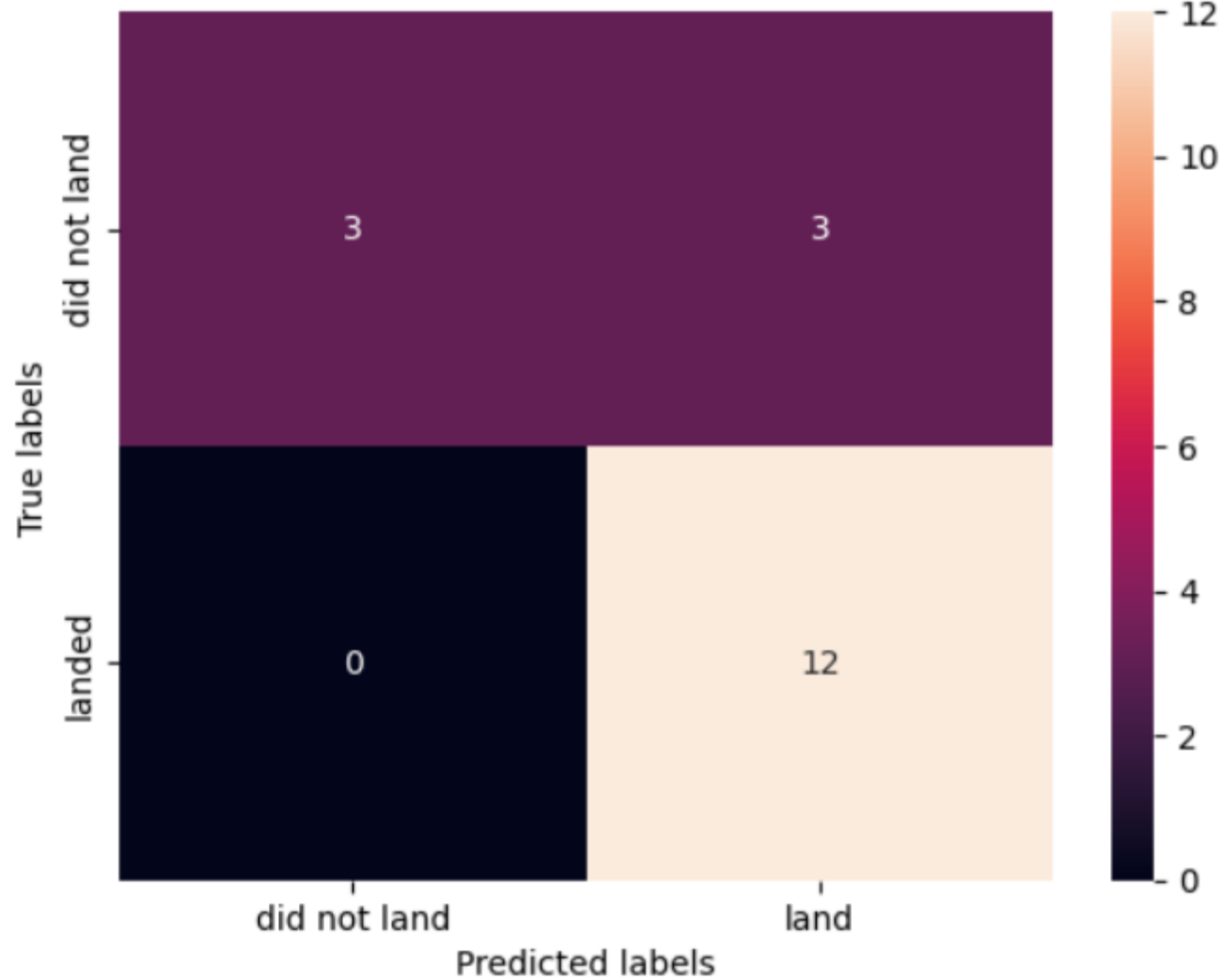
```
%sql SELECT Bo
```

```
* sqlite:///my_
Done.
```

Booster_Version

Listing the
success in
greater th

Confusion Matrix



payload

```
ND PAYLOAD_MASS__KG_ <
```

Total number of success

```
%sql SELECT Mission_Outcome, COUNT(*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Booster versions carried the max payload mass

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

Lists the records months failure landing outcome in drone ship

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql SELECT substr(Date,6,2) AS Month, "Landing_Outcome", Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,0,5)='2015' AND "Land:"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
-------	-----------------	-----------------	-------------

01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
----	----------------------	---------------	-------------

04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
----	----------------------	---------------	-------------

Rank the count of landing outcomes

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql SELECT "Landing_Outcome", COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_C
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

FOLIUM-MAP

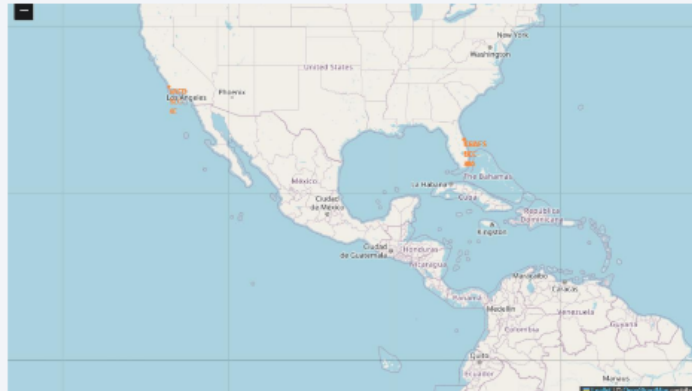
Mark all launch sites on a map

1. Are all launch sites in proximity to the Equator line?

- No, not all launch sites are in close proximity to the Equator.
- The launch site at Vandenberg Air Force Base (VAFB SLC-4E) is located at a latitude of 34.63, which is further from the Equator compared to the other sites in Florida.

2. Are all launch sites in very close proximity to the coast?

- Yes, all launch sites are in close proximity to the coast.
- The Cape Canaveral sites (CCAFS LC-40 and CCAFS SLC-40) and Kennedy Space Center (KSC LC-39A) are near the coast in Florida.
- Vandenberg Air Force Base (VAFB SLC-4E) is also near the coast in California.



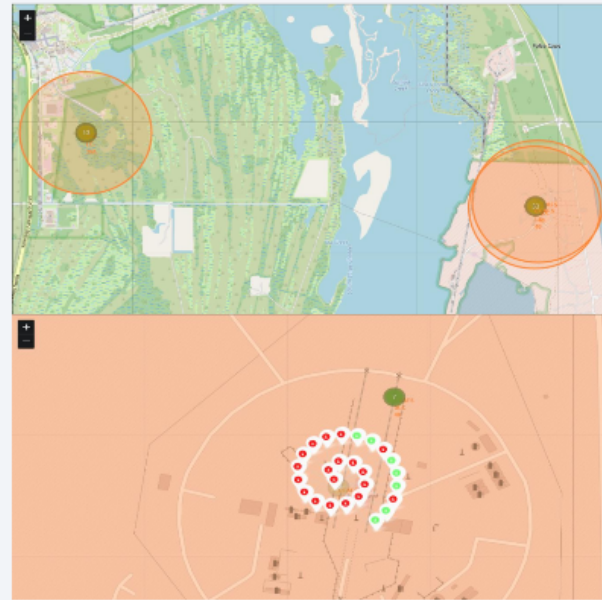
Most of the launch sites are in proximity to the equator line
The land is moving faster at the equator than any other place on the surface of the Earth

GIT HUB URL

https://github.com/JuanSer10/datasciencecoursera/blob/main/5%20lab_jupyter_launch_site_location.ipynb

Mark the success /failed launches

- This enhanced visualization with clustered markers allows for better exploration and analysis of SpaceX launch data. The clustering makes it easier to manage a large number of markers and observe patterns that might be hidden in a less organized plot. By examining the marker colors and popup information, you can gain deeper insights into the characteristics and distribution of SpaceX launches.
- For example, in the provided screenshot, out of 26 launch sites for CCAFS LC-40, there are 19 red markers and 7 green markers. This color-coding helps to quickly identify the success rate and other categorical distinctions of the launches from this specific site. The red markers might represent unsuccessful launches, while the green markers indicate successful ones, providing immediate visual feedback on the performance of launches at each site.



Explanation

From the colour labeled markers we should be able to easily identify which launch site have relatively high success rate

Green marker = successful

Red marker = Failed launch

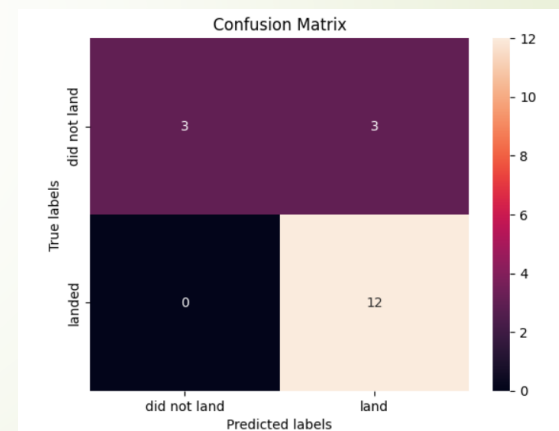
Predictive analysis

Classification accuracy

- Explanation
- Based on the scores of the Test set We can not confirm which method perform best
- Therefore, we tested all Methods based on the whole data set

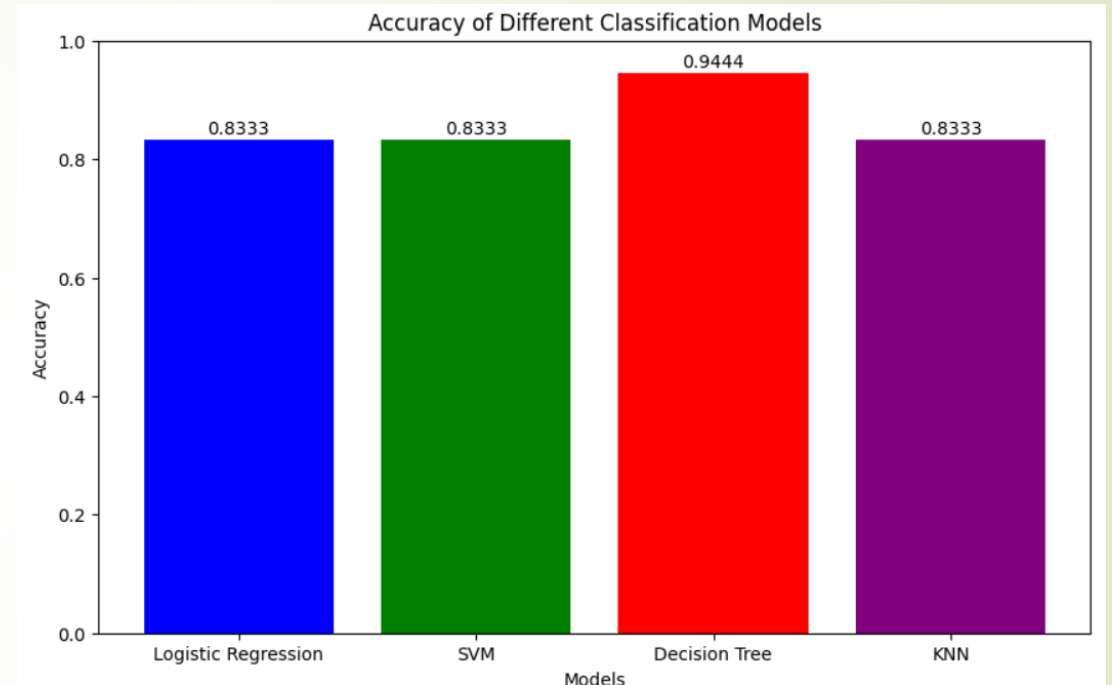
```
3]: best_tree = tree_cv.best_estimator_  
accuracy = best_tree.score(X_test, Y_test)  
  
print(f'Accuracy on test set: {accuracy:.4f}')
```

Accuracy on test set: 0.8333



Classification accuracy

Based on the results, the Decision tree is the highest model with an accuracy of 0,9444.





Conclusion



- Our analysis revealed that the "CCAFS LC-40" launch site has the highest success rate among all sites, accounting for 43.7% of successful launches. This indicates that this site might have optimal conditions or processes that contribute to a higher success rate.
- The scatter plot analysis showed that the "FT" booster version has a high success rate across various payload masses, demonstrating its reliability and robustness compared to other booster versions. This suggests that future missions might benefit from utilizing this booster version for improved success rates.